Q2. Loading and Inspecting the Iris Dataset

Original Aim: To load the Iris dataset into a Pandas DataFrame and inspect its structure. Display the first five records, number of rows and columns, column names, data types and summary statistics. Identify the numerical attributes and the target column, Species.

Aim

Load Iris into a Pandas DataFrame and inspect its records, dimensions, names, types, summaries, numerical attributes and target.

Requirements

Python, Pandas and scikit-learn. References: [D2], [D3].

Essential theory

A DataFrame is a labelled table of observations (rows) and variables (columns). A Series is one labelled column. The four measurements are numerical features; Species is a categorical classification target.

Step-by-step procedure

1. Load Iris, name its four measurements and map target codes to Species labels.
2. Display head(), shape, columns and dtypes.
3. Summarize measurements with describe(), count species and identify feature/target roles.

Complete Python program

```
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
features = [
    "SepalLengthCm", "SepalWidthCm",
    "PetalLengthCm", "PetalWidthCm"
]
species = ["Iris-" + name for name in iris.target_names]
df = pd.DataFrame(iris.data, columns=features)
df["Species"] = pd.Categorical.from_codes(
    iris.target, categories=species
)

print("FIRST FIVE RECORDS")
print(df.head().to_string(index=False))
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
print("DATA TYPES")
print(df.dtypes.to_string())
print("NUMERICAL SUMMARY")
print(df[features].describe().round(4).to_string())
print("Numerical features:", features)
print("Categorical target: Species")
print("SPECIES COUNTS")
print(df["Species"].value_counts(sort=False).to_string())
```

Code explanation

- `head()` displays the first five rows, not a random sample.
- `shape` returns (rows, columns); it is an attribute, not shape().
- `columns.tolist()` gives the names; `dtypes` reports each column's stored type.

- `describe()` returns count, mean, sample standard deviation, extremes and quartiles; 50% is the median.
- `value_counts(sort=False)` confirms class sizes. The displayed index is not an extra feature.

Verified output

```
FIRST FIVE RECORDS
 SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm     Species
           5.1           3.5            1.4           0.2 Iris-setosa
           4.9           3.0            1.4           0.2 Iris-setosa
           4.7           3.2            1.3           0.2 Iris-setosa
           4.6           3.1            1.5           0.2 Iris-setosa
           5.0           3.6            1.4           0.2 Iris-setosa
Shape: (150, 5)
DATA TYPES
SepalLengthCm     float64
SepalWidthCm      float64
PetalLengthCm     float64
PetalWidthCm      float64
Species          category
NUMERICAL SUMMARY
       SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
count       150.0000      150.0000       150.0000      150.0000
mean          5.8433        3.0573         3.7580        1.1993
std           0.8281        0.4359         1.7653        0.7622
min           4.3000        2.0000         1.0000        0.1000
25%           5.1000        2.8000         1.6000        0.3000
50%           5.8000        3.0000         4.3500        1.3000
75%           6.4000        3.3000         5.1000        1.8000
max           7.9000        4.4000         6.9000        2.5000
Categorical target: Species
SPECIES COUNTS
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
```

Observation

There are 150 rows, four float64 measurements and one categorical label. Numerical counts of 150 confirm no numerical gaps. Each species has 50 observations. Petal length has greater standard deviation than sepal length (see Q7).

Result

The structure, summaries and classification target were verified. Q10 instead uses PetalLengthCm as its numerical target.

Common errors and troubleshooting

- Six columns: a CSV may include Id; it is not a flower measurement.
- Species labels appear as 0/1/2: add the explicit target-to-label mapping.
- Species omitted by describe(): use value_counts() for category counts.

Viva questions and answers

- What is the difference between a Series and DataFrame? A Series is one labelled dimension; a DataFrame is a table of columns.
- What does shape return here? (150, 5), including Species.
- What does 50% in describe() mean? The median of the numerical column.
- Are species codes continuous measurements? No; they represent category identities, not distances between species.